



AI PET / MRI FUSION

FusionMap

PET-MRI precision from the scanners hospitals
already own

Team Code Blooded

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Real FusionMap output: fused PET/MRI, slice by slice

THE PROBLEM IN ONE NUMBER

13 mm

off target

That's how far the tumour glow on this patient's PET lands from the real tumour when it is simply overlaid on an MRI from a different scanner. Plan radiotherapy on that, and the beam treats healthy brain while the tumour escapes.

Measured, not estimated: across 12 test patients the naive overlay is off by 11.8 mm on average, up to 15 mm (95th percentile).

NAIVE OVERLAY



TUMOUR POSITION ERROR

13.2 mm

outline = true tumour

colour = PET uptake

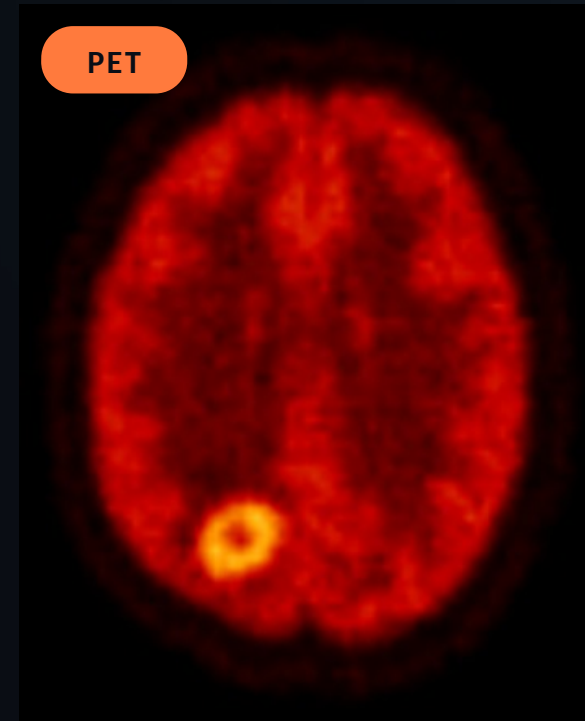
WHY FUSION

MRI sees two lesions. PET knows which one is alive



Shows WHERE

1 mm anatomy and soft tissue. But a dead lesion and a living one can look the same.



Shows WHAT'S ALIVE

Lights up tumour metabolism. But it is blurry (6 mm) and comes from another scanner.

WHY FUSION

Fused, they show exactly where the living tumour is

Dead tissue (radionecrosis)

MRI: looks like tumour. PET: cold, so it stays grey

Living tumour

The PET glow sits exactly on its MRI lesion



THE CATCH

\$4-6 M

for one integrated PET-MRI scanner

~10

of them in all of India

THE GAP

The scanners are already there. The fusion isn't



700+

cancer centres already own both a PET and an MRI scanner



6-8 lakh

patients a year who could benefit from fused imaging



~10

integrated PET-MRI scanners in the whole country



Today: the two scans sit on two screens and are compared by eye. Overlaid as-is, they miss by **11.8 mm** on average.

Centre, scanner and patient counts: team market research. Overlay error: FusionMap benchmark, 12 test patients.



FusionMap fuses them in three steps, in software



DICOM in from any PET and any MRI → **standard DICOM out** into the hospital. No new hardware.

Fix the mismatch between scanners



11.8 mm → **0.86 mm**

average PET-to-MRI alignment error, 12 held-out test patients

13×

smaller error than
overlaying the scans as-is

< 1 s

to align a full study on an
ordinary CPU

12 / 12

patients with 95 % of the
brain within 2 mm

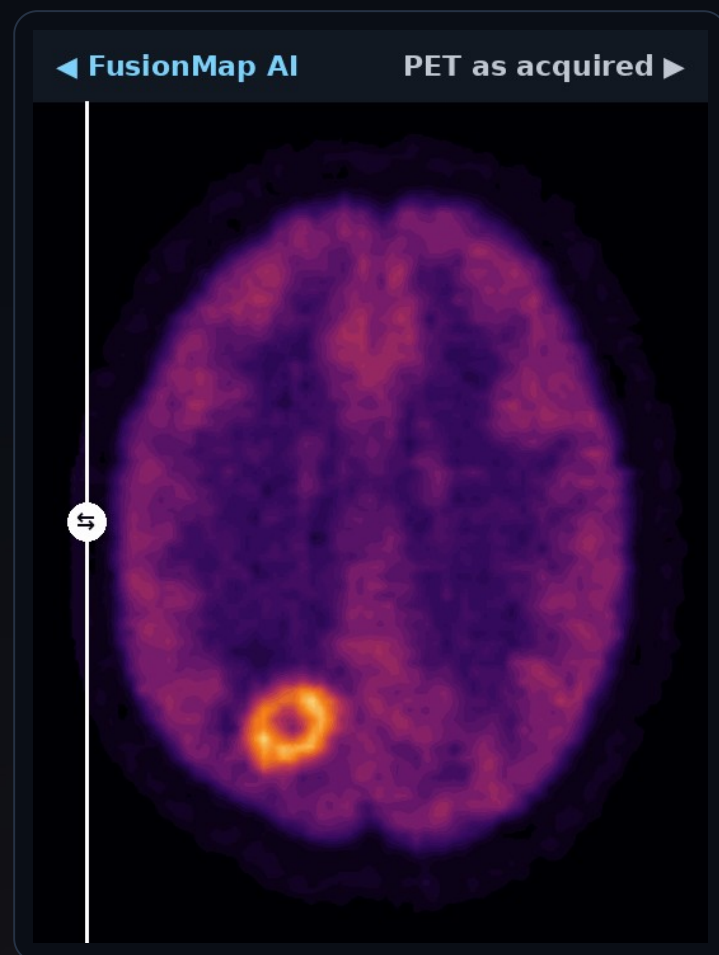


Mutual information aligns scans whose contrasts look nothing alike: PET versus MRI.



VoxelMorph AI is built in for organs that deform, and switches on only when its own validation beats the classical method.

Recover the detail scanner blur hides



36.6 dB

image fidelity (PSNR), up from 26.4 dB

0.98

structural similarity (SSIM), up from 0.65

99 %

of true tumour brightness recovered, up from 91 %

HOW THE AI WORKS



2.45 M parameters · trained on 29 simulated volumes · ~6 s per study on CPU (ONNX Runtime)



MRI-guided: the anatomy tells the network where the true edges are.



Never invents tumours: dead tissue that MRI shows but PET doesn't stays cold.

One colour-coded study, straight into PACS



WRITES STANDARD DICOM

- MR reference series
- PET in SUV, quantitative
- AI-enhanced PET
- Colour fusion image
- RT-STRUCT tumour contours



STRAIGHT INTO



Hospital PACS
DICOM C-STORE push



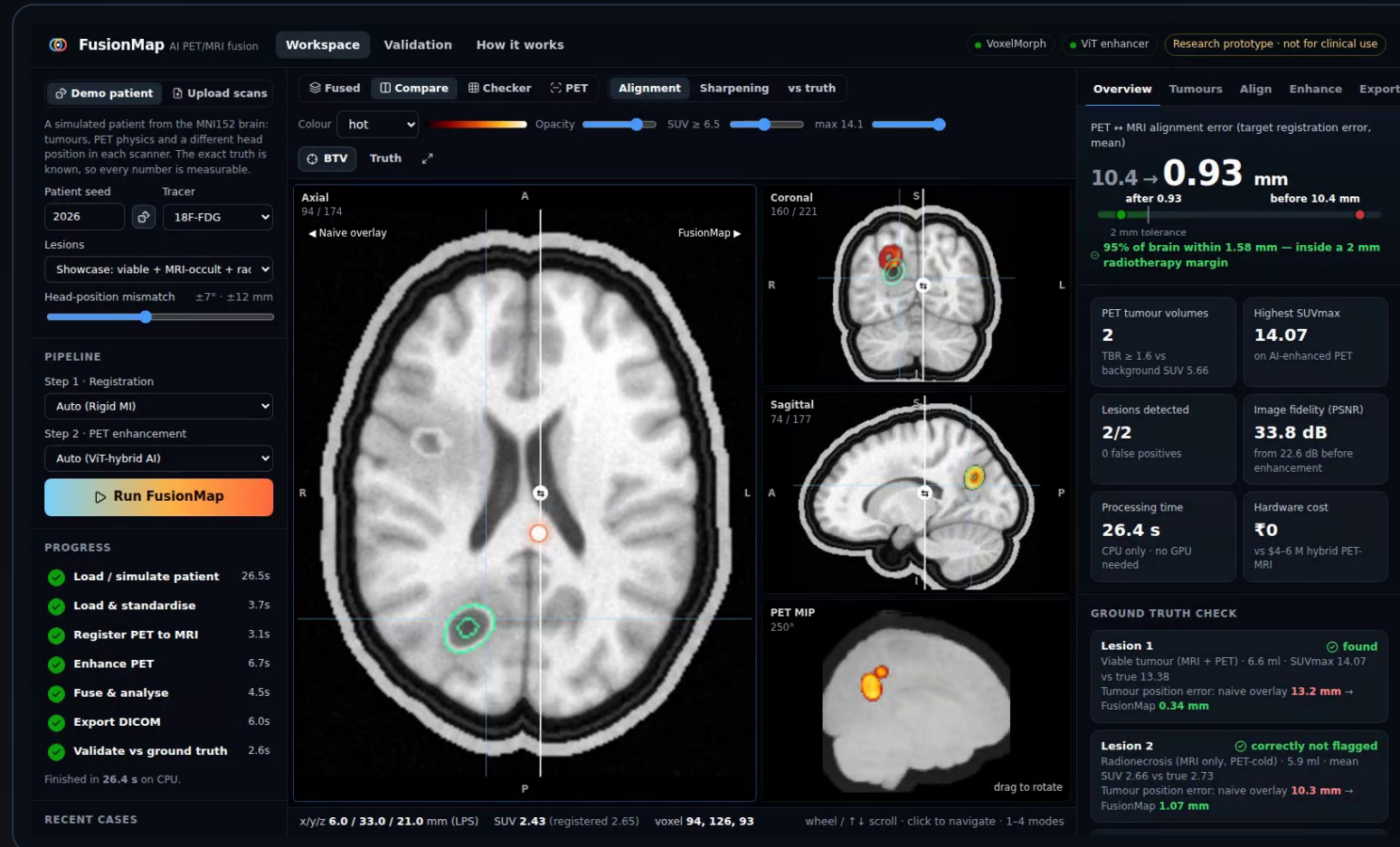
Radiotherapy planning
imports the RT-STRUCT



Printable report
for the tumour board

Automatic tumour analytics: SUVmax · SUVpeak · metabolic volume · TLG · tumour-to-background ratio. Every series shares one frame of reference, so any viewer lines them up.

See it run on a full study in 26 seconds



WHAT TO WATCH

- Alignment**
drag the swipe: naive vs FusionMap
- Sharpening**
PET as acquired vs AI-enhanced
- Ground-truth check**
every lesion verified against the truth
- Tumour metrics**
click a tumour: SUVmax, MTV, TBR
- Send to PACS**
plus the printable report

VALIDATION

Measured on patients where we know the truth

Alignment error, mm

lower is better · auto mode picks rigid MI

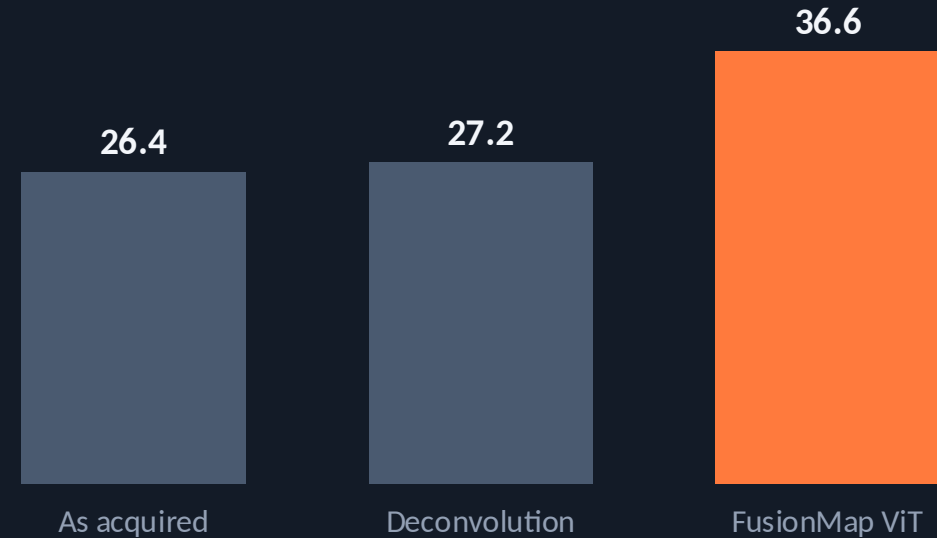
-93 %



PET image fidelity, PSNR in dB

higher is better

+10 dB



12 held-out test patients

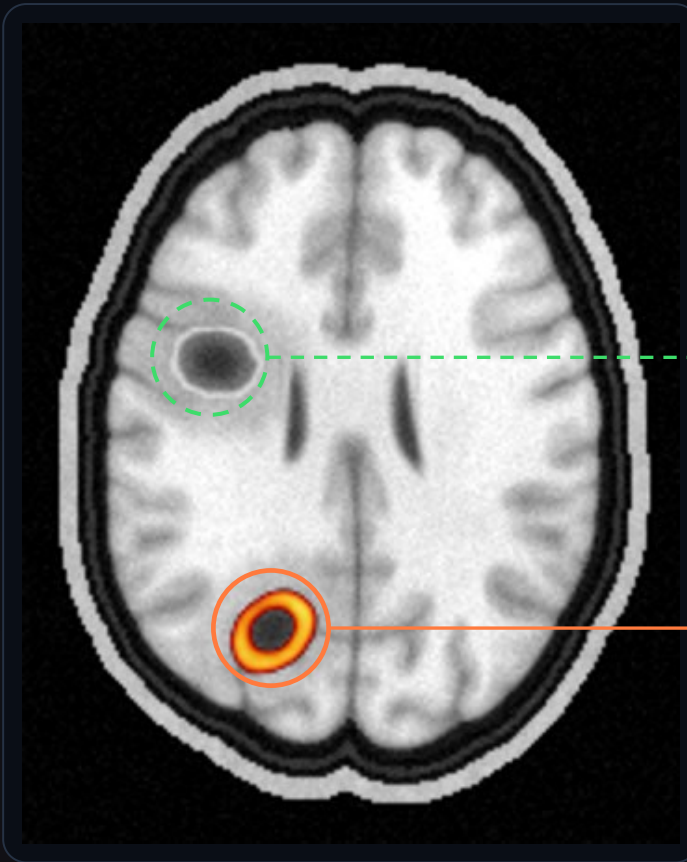
100 % of tumours detected

0 false positives

1 mm exact ground-truth grid



Built to be trusted, not just to impress



Radionecrosis

MRI says tumour, PET says dead: correctly not flagged

Living tumour

found, 0.3 mm from the truth



The radionecrosis trap

Our test patients include lesions that MRI shows but PET says are dead. An AI that copies the MRI would light them up. FusionMap keeps them cold: uptake ratio 1.05, where 1.0 is perfect.



An AI gate, not AI everywhere

Our VoxelMorph network matched classical alignment (1.172 vs 1.173 mm) but did not beat it, so auto mode keeps the classical method. AI switches on only when its own validation wins, and the raw PET is always exported too.

Runs on the hospital's own computer



PET-MRI-grade precision for the price of software



Integrated PET-MRI scanner

\$4–6 M

per scanner

New suite

shielding, siting, specialist staff

~10

in all of India



FusionMap

₹40–60 k

to build

₹0

new hardware: uses the PET and MRI already there

700+

centres that could run it



Patients

radiotherapy aimed at the living tumour



Doctors

one fused image instead of two screens



Hospitals

a new capability with no capital cost



From prototype to the clinic

NOW



NEXT



THEN



SCALE



Working prototype

Validated on 12 digital patients, DICOM in and out, PACS push, 45 tests

Real scans

Fine-tune on paired PET/MRI from The Cancer Imaging Archive (TCIA)

Reader study

Nuclear-medicine physicians compare FusionMap with hybrid PET-MRI

Clinic and body

CDSCO medical-device software pathway; lung and prostate with VoxelMorph



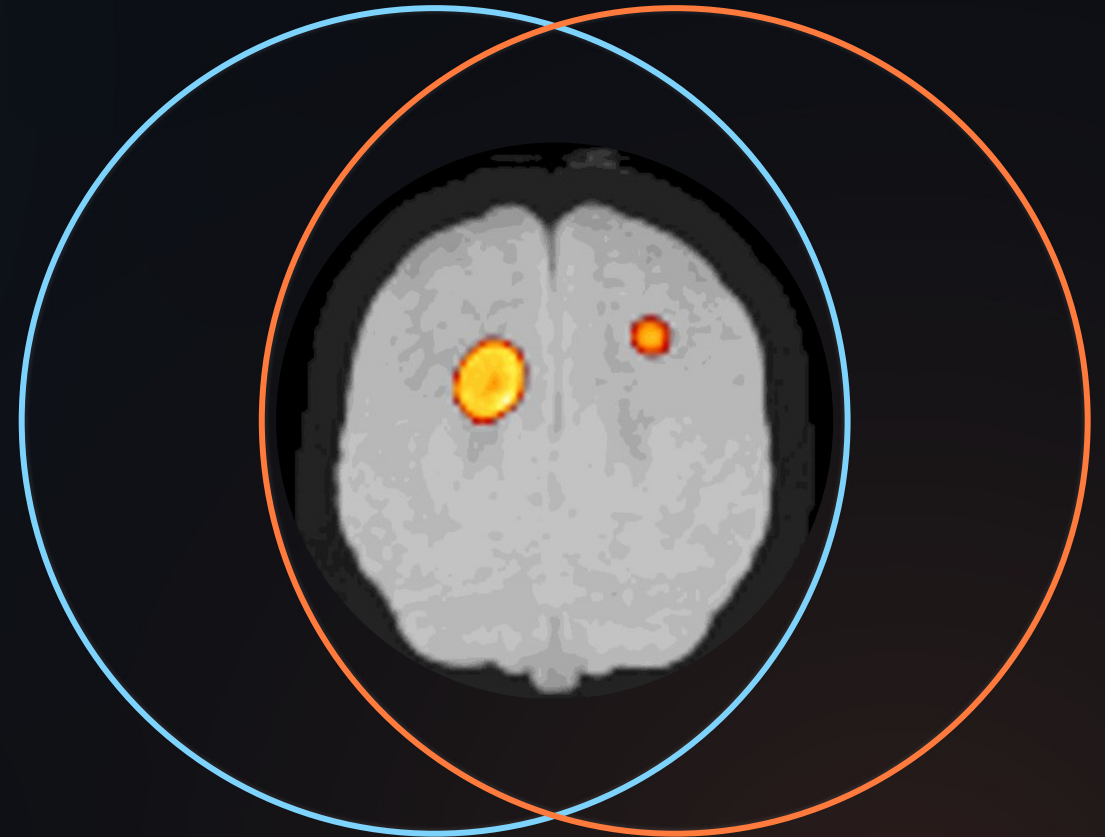
What would speed us up: a clinical partner with a PET and an MRI, for real-patient validation and the reader study.

THANK YOU • QUESTIONS?

700+ hospitals already
own a PET and an MRI.
**FusionMap turns them
into a PET-MRI.**



Team Code Blooded • Atharva Jadhav • Anushka Ghodekar • Rohan Leo
github.com/dev-parthdeshmukh/fusiontech



Questions we expect, answered

Validated on real patients?

Not yet, and we say so on every screen. Real scan pairs have no ground truth, so we use digital patients with exact answers. Next: TCIA data and a reader study.

Why not just use PET-CT?

CT shows brain soft tissue poorly and adds radiation. PET/MRI can cut the dose to about 20 % of PET/CT (Nensa 2014).

Does the AI copy the MRI?

That's exactly what we test for: radionecrosis stays cold (ratio 1.05), and the raw PET is always exported beside the AI PET.

Why a ViT, not CycleGAN?

With paired data, a supervised ViT hybrid is more faithful. We ship a CycleGAN trainer for hospitals that only have unpaired data.

Does it need a GPU or the cloud?

No. ONNX Runtime on a CPU, about 26 s per study, and patient data never leaves the hospital.

How does it fit the workflow?

DICOM in, standard DICOM out: MR, SUV PET, AI PET, colour fusion and RT-STRUCT, pushed to PACS by C-STORE.